STM32 Temperature Data Logger

Comprehensive Codebase, Hardware & Architecture Guide

Target Hardware: STM32F-Series (CBT6, e.g. STM32F103CBT6)	**Sensor:** DHT11 Digital Sensor (PA1)
Display: 16x2 HD44780 LCD via PCF8574 I2C (PB6/PB7)	**Storage:** W25QXX SPI NOR Flash (PA4..PA7)
Clock Frequency: 72 MHz (HSE 8MHz + PLL x9)	**Telemetry:** USART1 (115200 Baud 8-N-1)

1. System Overview & Working Principle

The STM32 Temperature Data Logger is an autonomous embedded monitoring instrument designed around the STM32F-series microcontroller (CBT6 package). It performs three core operations continuously: accurate ambient sensing, real-time visual feedback, and persistent non-volatile data logging.

Operational Sequence:

- **Live Sampling (Every 3 seconds):** The microcontroller requests a fresh temperature and relative humidity reading from the DHT11 sensor via a single-wire digital protocol. The reading is validated through an 8-bit checksum and immediately refreshed on the 16x2 I2C LCD along with heartbeat status indicator.
- **Periodic Logging (Every 5 minutes):** Every 300 seconds (300,000 ms), the reading is formatted into a 16-byte packed binary structure with header magic and verification checksum, then written to the external Winbond W25QXX SPI Flash memory.
- **Non-Volatile Retention:** Recorded history is preserved across complete power shutdowns. Upon system reboot, the storage engine mounts existing records without erasing historical data.
- **On-Demand Extraction:** Historical records can be extracted at any time through the USART1 serial port in comma-separated values (CSV) format using interactive terminal commands or the provided automated PowerShell/Python tools.

2. Hardware Pinout & Interfacing Matrix

The table below details all physical pin assignments, electrical characteristics, and interfacing notes for the STM32F CBT6 microcontroller:

Peripheral	Pin	STM32 Port	Mode / Electrical Specification
DHT11 Sensor	DATA	**PA1**	Single-wire bidir. **Requires 4.7k pull-up to 3.3V**
	VCC / GND	3.3V / GND	3.3V supply ensures safe logic level for PA1
16x2 I2C LCD	SCL	**PB6**	I2C1_SCL (100 kHz standard mode, 5V-tolerant)
	SDA	**PB7**	I2C1_SDA (PCF8574 backpack, 5V-tolerant)
	VCC / GND	5V / GND	5V ensures high contrast on HD44780 glass
W25QXX Flash	CS / NSS	**PA4**	GPIO Output Push-Pull (Chip Select)

	SCK / MISO / MOSI	**PA5 / PA6 / PA7**	SPI1 full-duplex master (Clock, MISO, MOSI)
	VCC / GND	3.3V / GND	**3.3V ONLY.** Never connect 5V to W25QXX chips!
ST-LINK V2	SWCLK / SWDIO	**PA14 / PA13**	Serial Wire Debug programming & debug port
UART Serial	TX / RX	**PA9 / PA10**	USART1 (115200 Baud, 8-N-1) for serial telemetry & CSV
Status LED	LED	**PC13**	Onboard heartbeat LED (Active-LOW)

3. Codebase Architecture & Modular Layout

The project has been refactored into a modular, decoupled hierarchy with separate header and source layers:

4. Detailed Module Implementation Analysis

A. Main Application Superloop (codes/main.c)

Role: Manages system initialization, invokes the boot sequence engine, periodic multi-rate execution, non-blocking timing, and UART user command interpretation.

• **Boot Sequence Integration:** Calls BootSequence_Run() before starting normal operations. If any critical subsystem fails, execution halts in an error loop with a 2 Hz LED alert.

• **Dual-Rate Timing:** Uses HAL_GetTick() state tracking rather than blocking delays. Task 1 refreshes sensor data on LCD every 3,000 ms; Task 2 appends records to W25QXX Flash every 300,000 ms (5 minutes).

• **Interactive UART CLI:** Listens via HAL_UART_Receive_IT() on USART1. Supports single-character interactive control: 'D' (dump CSV), 'S' (status report), 'L' (instant log), and 'E' (partition erase).

B. Professional Boot & Hardware Self-Check Engine (codes/boot_sequence.c & library/boot_sequence.h)

Role: Executes an 8-stage self-diagnostic sequence on startup within 2–5 seconds with visual feedback on the 16x2 LCD.

• **Stage 1: Power-Up:** Displays TEMP LOGGER / INITIALIZING during system voltage stabilization (500 ms).

• **Stage 2: System Check:** Displays SYSTEM CHECK / PLEASE WAIT as diagnostic routines begin (400 ms).

• **Stage 3: DHT11 Sensor Test:** Pings DHT11 over PA1. Displays DHT11 OK on success. If failed (non-critical), displays DHT11 FAIL / CONTINUING... and boots gracefully.

• **Stage 4: I2C LCD Handshake:** Validates I2C1 communication with PCF8574 backpack. Displays CHECK I2C LCD / I2C DISPLAY OK (critical).

• **Stage 5: W25QXX Detection:** Reads 3-byte JEDEC ID over SPI1 (PA4..PA7). Displays CHECK W25QXX / FLASH OK (critical).

• **Stage 6: Non-Destructive Memory Test:** Reads test signature from reserved Sector 0 (outside log sectors) and validates status register busy flag without erasing existing logs. Displays MEMORY TEST / PASS.

• **Stage 7: System Ready:** Displays SYSTEM READY / STARTING... if all critical checks pass, or BOOT ERROR / CHECK SYSTEM if critical components fail.

• **Stage 8: Normal Screen:** Seamlessly transitions to temperature monitoring screen: TEMP: 27°C / LOGGER: READY.

C. DHT11 Sensor Driver (codes/dht11.c & library/dht11.h)

Role: Executes single-wire bidirectional digital timing for temperature and humidity measurements.
• **DWT Cycle Counter Precision:** Uses ARM Cortex-M DWT->CYCCNT at 72 MHz (13.88 ns per tick) to measure microsecond-level pulse widths accurately without triggering timer interrupt overhead.
• **Protocol Stages:** 18 ms host LOW start pulse $\rightarrow$ 30 μs HIGH release $\rightarrow$ sensor presence handshake (80 μs LOW + 80 μs HIGH) $\rightarrow$ 40-bit data stream parsing.
• **Bit Discrimination:** 50 μs LOW followed by 26–28 μs HIGH is a binary '0'; 70 μs HIGH is a binary '1'. An 8-bit checksum confirms data integrity.

D. I2C 16x2 Character LCD Driver (codes/i2c_lcd.c & library/i2c_lcd.h)

Role: Drives standard HD44780 LCD panels over I2C using a PCF8574 8-bit GPIO expander.
• **4-Bit Nibble Protocol:** Sends bytes as two sequential 4-bit nibbles with Enable (EN) pulse strobing.
• **Auto-Address Probing:** Automatically tests address 0x27 (0x4E 8-bit) and 0x3F (0x7E 8-bit) on boot, automatically adapting to different PCF8574 chip variants.
• **Custom CGRAM Glyphs:** Defines a custom 5x8 degree symbol (°) in LCD CGRAM address 0 for clean thermal formatting.

E. W25QXX Universal SPI Flash Storage (codes/w25q128.c & library/w25q128.h)

Role: High-capacity non-volatile persistent storage for temperature logs across power resets.
• **Universal W25QXX JEDEC Detection:** Reads 3-byte JEDEC ID (`0x9F`). Accepts Winbond (`0xEF`) and compatible chips (`0xC8`), dynamically calculating total capacity: capacity_bytes = 1UL << cap (W25Q16 = 2MB, W25Q32 = 4MB, W25Q64 = 8MB, W25Q128 = 16MB).
• **Structured 16-Byte Packed Binary Record:**

Offset	Field	Type	Description
0..3	magic	uint32_t	Header synchronization marker (0xAA55A55A)
4..7	timestamp_sec	uint32_t	System uptime seconds at time of logging
8..9	temp_c	int16_t	Temperature in whole degrees Celsius
10..11	humidity	int16_t	Relative humidity in percentage (%)
12..13	record_id	uint16_t	Monotonically increasing sequential index
14..15	checksum	uint16_t	XOR checksum verifying record validity

5. Automation Scripts & Development Tooling

Five specialized PowerShell scripts located in the project root provide automated hardware diagnostics, compilation, ST-LINK programming, driver setup, and data extraction:

Script	Command Line	Function & Capabilities
check_environment.ps1	.\check_environment.ps1	Scans host PC for STM32CubeIDE, GCC ARM compiler, programmer CLI, active COM ports, and ST-Link drivers.
flash_firmware.ps1	.\flash_firmware.ps1	Locates STM32_Programmer_CLI and flashes target in SWD Hotplug mode without NRST lockup.
upload.ps1	.\upload.ps1	One-click script: builds source code from codes/ with headers in library/ and flashes via ST-LINK.
install_stlink_driver.ps1	.\install_stlink_driver.ps1	Extracts bundled stsw-link009.zip and launches 64-bit installer (dpinst_amd64.exe).
dump_logs.ps1	.\dump_logs.ps1 -Port COM3	Connects over UART, sends 'D' command to download all Flash records, and saves them to a timestamped CSV file.

6. Hardware Verification & Troubleshooting Runbook

- **DHT11 Returns Error (Code 1 or 2):** Check that the external 4.7 kΩ pull-up resistor is firmly connected between PA1 (DATA) and 3.3V. Ensure wire length is under 1 meter to avoid line capacitance degradation.

- **LCD Displays Black / White Solid Blocks:** Rotate the small blue contrast trimmer potentiometer on the rear of the PCF8574 I2C backpack with a small screwdriver until text is crisp.
- **W25QXX Flash Initialization Error:** Verify SPI CS pin is PA4 and VCC is strictly 3.3V (5V will damage or disable NOR Flash memory). Check that MISO (PA6) and MOSI (PA7) are not swapped.
- **ST-LINK Not Recognized:** Run .\install_stlink_driver.ps1 to install the official STSW-LINK009 USB driver package.